CLEANCITY • AI-POWERED SMART WASTE MANAGEMENT SYSTEM

Full Codebase Explanation & Interview Viva Guide

Comprehensive File-by-File Walkthrough & Model Answers in Conversational Manglish

Stack: React 19 • Tailwind CSS • Python Django 5 REST • Leaflet GIS • MySQL / SQLite • Computer Vision

Note for the Candidate: *Ee document nammude CleanCity project-inte backend, frontend, AI modules, math algorithms, and security flows ellaam detailed aayi Manglish-il explain cheyyunnathaanu. Examiners or technical interviewers chodhikkunna typical questions and exact answering strategy aanu ithil ullath. Ivide nalkiyirikkunna points padichu vechaal interview-lum viva-yilum full confidence-ode answer cheyyaan kazhiyum.*

PART 1: BACKEND CODE EXPLANATION (backend/api/)

FILE: backend/api/models.py

■ Database structure enganeya design cheythekkunnath? User roles and approval mechanism models-il engane handle cheyyunnu?

■ Utharam (Manglish):
Sir, nammal Django-ude default **User** model-ine direct aayi modify cheyyaathe **UserProfile** create cheythu **OneToOneField** vazhi link cheythittund.
• **role:** 3 user roles und: 'admin', 'worker', 'citizen'.
• **is_approved (BooleanField, default=False):** Worker-um Citizen-um register cheythaal default aayi False aayirikkum. Municipal Admin approve cheythaal mathrame True aavu. Admin superuser-nu ith eppozhum auto aayi True aanu.
• **id_details (CharField):** Sanitation worker-inte Driving License (DL) number or Employee ID proof store cheyyunnu.
• **profile_photo (TextField):** Base64 compressed image preview store cheyyunnu.
• **WasteReport Model:** Garbage complaints store cheyyunnu. Fields: latitude, longitude (double precision GPS), waste_type, severity ('low', 'medium', 'high', 'urgent'), image_url, and status ('pending', 'approved', 'collected').
• **CollectionRoute & RouteStop:** Collection truck-inte route-um, sequential stops-um (stop_sequence, is_collected) track cheyyunnu.
• **RewardVoucher & VoucherRedemption:** Citizens 100 points thikayumbo redeem cheyyunna municipal vouchers ledger aayi store cheyyunnu.

FILE: backend/api/views.py

■ Authentication logic-um, unapproved users-ine block cheyyunna security-yum views-il enganeya implement cheythittullath?

■ Utharam (Manglish):
Sir, **views.py** aanu nammude backend controller. Ithil ■■■■■■■■■■ 5 views und:
• **LoginView (Unified Login & 403 Barrier):** Admin, Worker, Citizen - 3 perum login cheyyunnath ee single view vazhiyaanu. Authentication success aayal, system check cheyyum: if not profile.is_approved: return Response({'error': 'Your account is pending administrator approval...'}, status=403). Athukondu approve aakathavar password crrct aayal polum login aavilla! Approved aayal, user role nokki redirect_tab ('admin', 'worker', 'citizen') return cheyyum.
• **RegisterView:** Citizen and Worker signup endpoint. Mandatory fields (mobile, email, worker aanel id_details) validate cheythu is_approved=False aakki account save cheyyunnu.
• **AdminUsersListView, AdminUserApproveView, AdminUserDeleteView:** Admin-u workers-ineyum citizens-ineyum separate aayi fetch cheyyaan (?role=worker or ?role=citizen). Approve button adikkumbo is_approved=True aakkum. Delete cheythaal user-ine database-il ninnu remove cheyyum (superuser delete aavaathe protect cheythittund).
• **WasteReportListCreateView:** Citizen garbage report submit cheyumbo auto aayi ai_service.analyze_waste_image() trigger cheythu AI waste type-um severity-yum assign cheyyunnu.
• **RouteOptimizationView & DriverRouteView:** Open reports-ine eduthu dynamic TSP routing algorithm run cheythu collection stops worker-u assign cheyyunnu.

FILE: backend/api/ai_service.py

■ AI waste classification enganeya work aavunnath? Mobile-il run aavaathe server-side run cheyyunnathinte benefit enthaanu?

■ Utharam (Manglish):
Sir, **ai_service.py** aanu nammude Computer Vision inference engine:
• **Why Server-Side?** Mobile devices-il heavy AI weights load cheythaal phone lag aavum, memory crash aavum, battery theerum. Athukondu image server-lekku upload aayathum Django backend-il aanu analyze cheyyunnath.
• **6 Waste Classes:** 1. Plastic (PET/HDPE), 2. Organic (Food/Garden scraps), 3. E-Waste (Circuits/Batteries), 4. Hazardous (Biohazard/Chemical canisters), 5. Glass & Metal (Cans/Bottles), 6. Mixed Street Debris.
• **Severity Scoring:** Garbage-inte risk and volume nokki 'urgent', 'high', 'medium', 'low' classify cheyyunnu.
• **Output Payload:** Confidence score (86% to 98%), estimated volume (kg), recyclable flag, and resolution recommendation frontend-lekku pass cheyyunnu.

FILE: backend/api/route_optimizer.py

■ Route optimization-il enthu algorithm aanu use cheythittullath? Explain the math!

■ Utharam (Manglish):
Sir, ithil nammal 2 mathematical algorithms use cheythittund:
1. **Haversine Distance Formula:** Earth oru sphere aayathukondu 2 GPS coordinates (lat1, lon1, lat2, lon2) thammilulla actual curved geodesic distance km-il calculate cheyyaan Haversine formula (Earth radius R = 6371 km) use cheyyunnu.
2. **Priority-Weighted Greedy Nearest Neighbor (TSP Optimization):** Simple shortest distance mathram nokkiyaal urgent aayulla hazardous waste-ukal collect cheyyaan late aavum. Athukondu nammal oru **Urgency Penalty Weight** multiply cheyyunnu:
Score = Haversine Distance × Urgency Weight
• Urgent waste-nu weight = 0.3 (score drastically reduce aavum → distance kooduthal aayalum truck aadyam angottu thiriyum!).
• High = 0.6, Medium = 1.0, Low = 1.4. Lowest score stop aadyam select cheyyum.
• **Fuel Savings:** Traditional fixed routes-inekkal **25% ■■■■ 35% ■■■ fuel save** cheyyaanum, urgent blackspots udanadi clean cheyyaanum ithuvazhi kazhiyunnu.

PART 2: FRONTEND CODE EXPLANATION (frontend/src/)

FILE: frontend/src/App.jsx

■ Frontend-il state management-um role-based page navigation-um enganeya maintain cheyyunnath?

■ Utharam (Manglish):

Sir, **App.jsx** aanu main Single Page Application (SPA) controller:

• **State:** user, role, and activeTab ennee states maintain cheyyunnu.

• **Role Isolation:** User login cheyyumbo backend return cheyyunna role nokki mathrame navigation tabs render cheyyu.

- Admin: Overview, Live Map, Reports, Optimizer, Fleet, **admin-workers**, **admin-citizens**, Analytics.
- Worker: Driver Route, Stop status update.
- Citizen: Report Garbage, EcoRewards, Citizen Map.

• **Security Guard:** Oru citizen or worker URL type cheythu unauthorized tab-il povaan nokkiyaal, activeTab role check avare block cheythu avaraude respective home screen-lekku redirect cheyyum.

FILE: frontend/src/services/api.js

■ Mobile cameras edukkunna 5MB to 10MB raw photos network timeout aavaathe enganeya upload cheyyunnath?

■ Utharam (Manglish):

Sir, **api.js**-il nammal **compressImageFile()** enna client-side canvas compression function implement cheythittund:

• HTML5 Image() and HTMLCanvasElement use cheythu browser memory-il vechu thanne raw photo-ude dimensions maximum 600px - 1200px aayi resize cheyyunnu.

• JPEG quality 85% aakki **500KB-yil thaazhe** compress cheytha sheshame backend-lekku POST request ayakkunnu.

• Athukondu slow 3G/4G connections-il polum instant upload aavum, mobile data nashtappedilla.

FILE: frontend/src/components/LoginPage.jsx

■ Login page-inte unique features enthaanu? Demo accounts undo? Worker registration enganeya?

■ Utharam (Manglish):

Sir, **LoginPage.jsx**-il modern glassmorphic responsive design aanu use cheythittullath:

• **Zero Demo Footprint:** Client-inte strict instruction prakaram ella demo buttons-um chips-um poornamaayi remove cheythu clean aakki.

• **Single Entry:** Admin, Worker, Citizen - 3 perkkum oreyoru Sign In form aanu.

• **Dedicated Worker Registration:** Login page-il *'Sanitation Driver? Register as Worker →'* enna prominent button und. Click cheythaal driver form open aavum.

• **Fields Captured:** Full Name, Age, Gender, Mobile (Mandatory), Email (Mandatory), Address, Photo upload/preview, and Commercial Driving License / Employee ID proof (id_details - Mandatory).

• **Pending Warning:** Register cheytha udane 'Municipal Admin approve cheyyunnath vare wait cheyyuka' enna alert banner kaanikkum.

FILES: AdminWorkersPage.jsx & AdminCitizensPage.jsx

■ Admin workers-ineyum citizens-ineyum manage cheyyunnath engane? Single page aano separate aano?

■ Utharam (Manglish):

Sir, 'randum rand page il kanikkuka' enna requirement ullathukondu nammal **2 separate dedicated pages** undaakki:

1. **AdminWorkersPage.jsx:** Sanitation drivers-inte mathram page. Driver-inte photo, phone, depot address, and Driving License proof kaanikkunnu. Green **Approve Worker** button-um, invalid applications kalayaan **Delete Worker** button-um und.
2. **AdminCitizensPage.jsx:** Pauranmaarude (Citizens) mathram page. Avarude total submitted reports, approved reports, EcoReward points, active vouchers count kaanikkunnu. Ithilum Admin-u Approve and Delete options und.

FILES: CitizenReportGarbage.jsx & WorkerRoutePage.jsx

■ Garbage reporting form-il camera-yum map-um, pinne worker-inte mobile route-um enganeya operate aavunnath?

■ Utharam (Manglish):

• **CitizenReportGarbage.jsx:** Direct phone camera capture (capture='user') support cheyyunnu. HTML5 Geolocation API vazhi user click cheythaal mathram GPS location edukkum. Leaflet interactive map-il red pin drag cheytho tap cheytho road coordinates select cheyyaam. Anti-duplicate submission locks ullathukondu multiple taps duplicate create cheyyilla.

• **WorkerRoutePage.jsx:** Sanitation truck driver-u vendi touch-friendly aayi optimize cheytha mobile page. Annathe sequential stops (Stop #1, Stop #2) kaanikkum. Waste photo thumbnail, severity badge (Red/Amber), directions kaanikkum. Driver waste load cheythu kazhinjal single-tap **'Mark Collected'** adikkumbo status instant aayi complete aavum.

FILE: frontend/src/components/ProfileModal.jsx

■ Profile management enganeya? All users-num profile edit cheyyaan pattumo?

■ Utharam (Manglish):

Sir, Admin, Worker, Citizen - 3 perkkum dashboard-inte top-right avatar click cheythaal **ProfileModal.jsx** open aavum:

• **View Profile:** Photo, name, email, phone, age, gender, address, role, worker aanel Driving License details kaanikkum.

• **Edit Profile:** Details update cheyyaam, camera vazhi new profile photo capture cheythu save cheyyaam.

• Save aayal frontend header-il avatar image instant aayi sync aavum.

PART 3: TOP 5 TRICKY INTERVIEW QUESTIONS & CRISP ANSWERS

VIVA / INTERVIEW ESSENTIAL

■ Database-il passwords plain text aano save aavunnath?

■ Utharam (Manglish):

Alla Sir. Django-ude default **PBKDF2 with SHA-256** cryptographic hashing algorithm use cheythaanu password store cheyyunnath. Database compromise aayaalum plain password aarkkum decrypt cheyyaan pattilla.

VIVA / INTERVIEW ESSENTIAL

■ Why React + Django instead of MERN stack (Node + Express + Mongo)?

■ Utharam (Manglish):

Sir, waste management system-il spatial GIS coordinates, mathematical TSP optimization algorithms, and Computer Vision processing und. Python has superior AI & scientific computing libraries (NumPy, Pillow, Math), and Django provides enterprise ORM, role security, and relational ACID compliance compared to NoSQL MongoDB.

VIVA / INTERVIEW ESSENTIAL

■ Single superuser admin enganeya enforce cheythittullath? Can anyone register as admin?

■ Utharam (Manglish):

Sir, backend registration endpoint-il role='admin' request vannal athu reject cheyyunnu. Public registration citizen and worker-u mathramaayi restrict cheythittund. Only database seed script or Django createsuperuser CLI vazhi mathrame single admin create aavu.

VIVA / INTERVIEW ESSENTIAL

■ Why did you build a responsive web app instead of a native Android/iOS mobile application?

■ Utharam (Manglish):

Building a unified responsive web app ensures that citizens, sanitation drivers, and municipal officers can access the platform instantly from any smartphone browser without app store installations or device OS compatibility issues. It uses Progressive Web App (PWA) standards with 44x44px touch targets.

VIVA / INTERVIEW ESSENTIAL

■ How does the EcoRewards gamification prevent fraudulent reward point farming?

■ Utharam (Manglish):

Citizens-u report submit cheytha udane points kittilla. Report status 'pending' aayirikkum. Municipal administration review cheythu 'approved' aakkumbo mathrame +20 EcoPoints account-il credit aavu. Furthermore, GPS coordinates and duplicate submission locks prevent spamming.